

Module 3: Paired multiome interpretation with MOFA

Instructional time: 30 min. Bloom: Analyze. Notebook: notebooks/03_multiome.ipynb

1. Overview

Module 3 interprets a fitted MOFA model on HuBMAP SNARE-seq2 data from Donor_2 (HBM828.GPVG.252). That multiome shares a block and a section with the Donor_2 10x snRNA teaching block used in Modules 1-2, so the two assays can be compared as a matched pair (different nuclei; no per-nucleus join). The lesson asks whether RNA and ATAC share latent structure, whether MOFA found cell identity at all, what the ATAC gene-activity view actually encodes, and whether a second integrator (WNN) does better.

Key components. [multiome_mofa.hdf5](#) as the default download; a committed Azimuth / WNN label cache in the package; variance and dominance tables; cluster concordance; QC association; factor versus label association; genome-wide and focus-gene bridges.

Learning objectives.

- Explain what paired multiome adds beyond Module 1/2 snRNA-seq, and what a matched block licenses.
- Read MOFA variance tables and recognize modality-private factors.
- Tell a broken model from an absent ATAC signal using cell-type and QC diagnostics.
- Separate ATAC gene activity from RNA expression when interpreting correlations.

Default compute stays inside the course RAM floor of 8 GB, with 16 GB recommended (`load_h5mu: false`). Learners do not download the ~14.7 GiB MuData file.

2. Before you begin

Prerequisites. Module 1 (QC objects) and Module 2 (integrated HuBMAP object / pseudobulk exports). Conda env `CFDE_lung_env` (or the pinned pip env). Default path uses the fitted MOFA HDF5 plus the committed label cache; full MuData is not a learner download.

Time. Instructional: 30 min. Compute is short on the default path. Measured peak RSS 7.1 GB; 16 GB recommended, 8 GB is the minimum.

Biological question

In Donor_2 SNARE-seq2 nuclei, do RNA and ATAC share latent structure in the shipped MOFA model? If not, did the method fail, or is the ATAC gene-activity view empty of cell-type information?

Data used here

File	Local path	Role
multiome_mofa.hdf5	data/source/HUBMAP/Donor_2/SNAREseq/HBM828.GPVG.252/multiome_mofa.hdf5	Default interpretation object
Label cache	outputs/tables/module3_label_cache_HBM828.GPVG.252.tsv	Committed Azimuth labels, WNN Leiden, WNN UMAP

secondary_analysis.h5mu (optional)	same SNARE directory	Full MuData if module3.params.load_h5mu: true (~14.7 GiB; not a learner download)
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Portal: [HBM828.GPVG.252](#). Modules 1-2 use 10x snRNA-seq on Donor_2 block HBM473.NKMR.872, which shares a block and a section with this multiome.

SNARE folders are named by the public processed dataset ID, not by a distinct primary block ID. That is the exception relative to the 10x snRNA teaching folders.

3. Methods

Parameters below are live values under `module3` in `config/paths.yaml`, collected in the Appendix. Scripts live in `scripts/nb05_multimodal/`. Default compute is interpretation-only from the shipped MOFA HDF5 plus the committed cache; full MuData is optional and is not required.

Each Methods step below corresponds to a section of `notebooks/03_multiome.ipynb`. The notebook is organized by the question each diagnostic answers rather than by step, so several Methods steps feed one notebook section:

Methods step	Notebook section
0. The MOFA model and variance explained per view	conceptual, no notebook section
1. Default path: interpret shipped MOFA	1. Object composition and barcode pairing
2. Barcode pairing check	1. Object composition and barcode pairing
3. Variance explained and factor dominance	2. Variance explained across modalities
4. Factor association with cell identity	3. Factor association with cell identity
5. Content of the ATAC gene-activity view	4. Content of the ATAC gene-activity view
6. Weighted nearest neighbor as a second integration	5. Weighted nearest neighbor concordance
7. Genome-wide RNA-ATAC gene bridge	6. Gene bridge and focus genes
8. Focus-gene correlations, global and within cell type	6. Gene bridge and focus genes

Notebook section 0 is environment setup, covered in Module 0.

0. The MOFA model and variance explained per view

MOFA (Multi-Omics Factor Analysis) is an unsupervised factor model. Given two or more matrices measured on the same cells, it learns a small set of latent factors and, for each factor, a set of weights per feature in each modality. A factor is a direction of variation; its weights say which genes or which chromatin features move along it.

The quantity that matters for interpretation is **variance explained per factor per view**. For each factor MOFA reports how much of the variance in the RNA view it accounts for, and how much of the variance in the ATAC view. That decomposition is what distinguishes the two outcomes this module asks you to tell apart:

- A **shared** factor explains a substantial share of variance in **both** views. It represents coordinated variation, where the same biological axis is visible in transcription and in chromatin accessibility.
- A **modality-private** factor explains variance in **one** view and close to none in the other. It represents structure confined to a single assay.

Applied to this dataset, RNA is a count matrix over genes and ATAC is summarized as gene-level accessibility, so both views are indexed by gene and a shared factor is biologically plausible: a gene whose promoter opens might be expected to rise in expression. Whether that coupling is actually present in these nuclei is the empirical question, and the variance decomposition answers it directly rather than by inference.

Two properties of the method are worth carrying into the interpretation. MOFA is unsupervised, so factors are not labeled by cell type or condition and any biological meaning has to be argued from the weights. And the number of factors is set in advance, so factors beyond the informative ones carry near-zero variance and should be read as absent rather than as weak effects.

1. Default path: interpret shipped MOFA

Script. `scripts/nb05_multimodal/load.py` (`multiome_paths`, `load_mofa`); inventory helpers in `scripts/nb05_multimodal/inventory.py`.

What. Load `multiome_mofa.hdf5` with views configured as `rna` and `atac_cbg` (`module3.modalities`). Do **not** refit MOFA in the lesson (`module3.params.interpretation_only: true`, `load_h5mu: false`). Optional ATAC bin view `atac_cbb` exists in config but `use_atac_bin` defaults false. The teaching product is HuBMAP SNARE-seq2 for Donor_2 (HBM828.GPVG.252), matched to the Donor_2 10x snRNA teaching block.

Parameters: see Appendix, section 1.

Why. Refitting is slow, non-identical across environments, and unnecessary for teaching how to read a DCC-provided multiome factor model. The HDF5 plus the committed cache keep the module inside the course download budget.

Modules 1-2 and 4 are unaffected unless the multiome file is part of the learner download manifest.

2. Barcode pairing check

Script. `scripts/nb05_multimodal/bridge.py` (`barcode_overlap_table`, `barcode_overlap_from_mofa`); optional MuData barcode sets in `scripts/nb05_multimodal/load.py` (`barcode_sets`).

What. Report RNA and ATAC barcode counts and overlap (`module3_barcode_overlap.tsv`). With MuData unloaded, overlap is asserted by MOFA's shared sample list rather than an independent MuData crosstab. State that basis explicitly in results.

Parameters: see Appendix, section 2.

Why. Paired multiome claims require shared nuclei. Stating the basis of the overlap avoids implying a stronger check than was run on the default path.

3. Variance explained and factor dominance

Script. `scripts/nb05_multimodal/bridge.py` (`mofa_factor_dominance`); variance export in `scripts/nb05_multimodal/export.py`; plots in `scripts/nb05_multimodal/plots.py` (`plot_mofa_variance`).

What. Summarize total R2 by view, per-factor R2, and dominance (which view owns each factor). Gate stored R2 units (percent vs fraction) before quoting totals. Config `module3.analysis.mofa_variance_max_factor` marks trailing factors as numerically dead on this object (note: this key lives under `'analysis'`, not `params`).

Parameters: see Appendix, section 3.

Why. Shared latent biology would show factors with substantial R2 in **both** views. Modality-private factors mean stacked single-modality structure. Dead trailing factors are a first-class result, not a plotting inconvenience. Misreading percent-scale R2 as fractions invents impossible totals.

4. Factor association with cell identity

Script. `scripts/nb05_multimodal/load.py (load_label_cache, assemble_nucleus_frame); scripts/nb05_multimodal/bridge.py (factor_label_association); scripts/nb05_multimodal/plots.py (plot_factor_label_heatmap).`

What. Join the committed cache (Azimuth labels, WNN Leiden, WNN UMAP) onto MOFA sample order. For every factor score, compute eta squared against Azimuth, WNN, RNA Leiden, and ATAC Leiden. Write `module3_factor_label_association.tsv`. Cache generation from the MuData file is an authoring step (`scripts/nb05_multimodal/cache.py`); learners read the committed TSV.

Parameters: see Appendix, section 4.

Why. A modality-private MOFA fit can still have found biology in one view. Eta squared against cell identity is how you tell "the method worked on RNA" from "the object is noise."

5. Content of the ATAC gene-activity view

Script. `scripts/nb05_multimodal/load.py (mofa_sample_metadata); scripts/nb05_multimodal/bridge.py (cluster_concordance, qc_association); scripts/nb05_multimodal/plots.py (plot_cluster_concordance, plot_qc_association).`

What. `MOFA samples_metadata/group1` already carries RNA Leiden, ATAC Leiden, ArchR Clusters, and the ATAC QC panel, so these diagnostics do not need the MuData file. Compute pairwise adjusted Rand index and normalized mutual information. Compute eta squared of QC metrics under each clustering, log10-transforming fragment and TSS count columns and recording that transform.

Parameters: see Appendix, section 5.

Why. If ATAC Leiden is chance-level against cell type and strongly associated with sequencing depth, the ATAC gene-activity matrix is not a legitimate basis for integration. That is a property of this representation, not a proof that chromatin is uninformative in lung.

6. Weighted nearest neighbor as a second integration

Script. `scripts/nb05_multimodal/bridge.py (cluster_concordance); scripts/nb05_multimodal/plots.py (plot_wnn_umap).`

What. The cache supplies WNN Leiden and WNN UMAP from the portal MuData object. Compare WNN to RNA Leiden, ATAC Leiden, and Azimuth. Plot the WNN embedding colored by Azimuth and by ATAC Leiden.

Parameters: see Appendix, section 6.

Why. A joint embedding can look successful while still leaning on one modality. WNN is the portal's second integrator; it is the control for "maybe MOFA was the wrong method."

7. Genome-wide RNA-ATAC gene bridge

Script. `scripts/nb05_multimodal/bridge.py (rna_atac_gene_bridge); metric configured as module3.analysis.bridge_metric.`

What. Join RNA and ATAC features by HUGO symbol on MOFA inputs; summarize dispersion relationships with metric `feature_std_on_mofa_inputs`. Report genome-wide Pearson/Spearman on those paired dispersions.

Parameters: see Appendix, section 7.

Why. ATAC gene activity is not RNA expression. Weak global correlation is an expected scientific result for this object, not a pipeline failure.

8. Focus-gene correlations, global and within cell type

Script. `scripts/nb05_multimodal/bridge.py` (`focus_gene_bridge`, `focus_gene_correlations`, `focus_gene_correlations_by_cell_type`, `combine_focus_correlations`, `focus_gene_availability`).

What. Correlate configured focus genes in RNA and ATAC spaces. Joining the cache supplies Azimuth labels, so within-cell-type rows are emitted on the default path. Config `module3.thresholds.min_nuclei_correlation: 50` flags correlations with `n` below this threshold. `focus_gene_cell_types` was re-checked against the Azimuth labels present in this `Donor_2` object. `focus_genes_not_in_both_modalities` (e.g. EPCAM) documents genes that cannot support a paired claim.

Parameters: see Appendix, section 8.

Why. Focus genes connect Module 3 to lung programs taught in Modules 1-2 without pretending ATAC gene activity equals RNA abundance. Explicit low-`n` and missing-modality rows prevent silent empty correlations.

MOFA inputs or cache changed.

4. Results

Object composition and barcode pairing

rna_n_barcodes	atac_n_barcodes	n_exact_overlap	overlap_fraction_of_rna	overlap_fraction_of_atac	paired_multio	interpretation
21978	21978	21978	1.0	1.0	True	NOT independent pairing evidence from a barcode crosstab. Pairing is asserted by MOFA file construction: one shared sample barcode list is written once for rna + atac_cbg views. <code>n_exact_overlap</code> equals that list length by definition (MuData barcode crosstab skipped because <code>load_h5mu=False</code>).

Source: `module3_barcode_overlap.tsv`

azimuth_label	n_nuclei	fraction
EC general capillary	8339	0.379425
AT1	2210	0.100555
EC arterial	2030	0.092365
Alveolar fibroblasts	1620	0.07371
CD8 T cells	1509	0.06866
Multiciliated (non-nasal)	1109	0.05046

AT2	849	0.03863
Pericytes	676	0.030758
Ionocyte	601	0.027346
EC venous pulmonary	523	0.023797
EC venous systemic	474	0.021567
CD4 T cells	441	0.020066
Non-classical monocytes	343	0.015607
EC aerocyte capillary	222	0.010101

Showing 14 of 34 rows. Source: module3_azimuth_composition.tsv

RNA and ATAC barcode overlap equals the shared MOFA sample list. That is construction, not an independent MuData crosstab: `load_h5mu` is false. The cache row count and Azimuth label count are stamped in `outputs/reports/module3_label_cache_provenance.json`. The matched 10x snRNA block is `HBM473.NKMR.872` (same block and section; different nuclei).

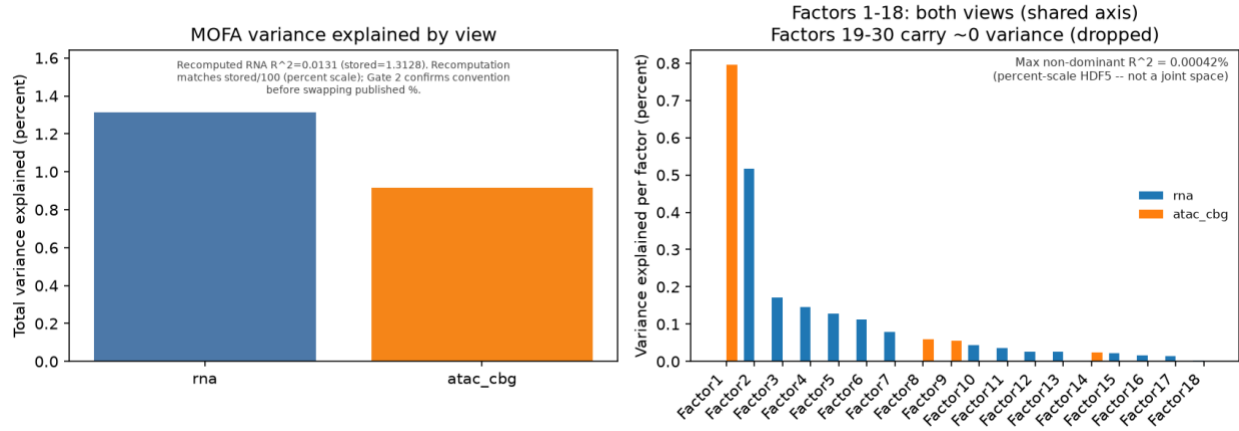
Variance explained across modalities

view	r2_total	units
rna	1.312837	percent
atac_cbg	0.914476	percent

Source: module3_mofa_variance_total.tsv

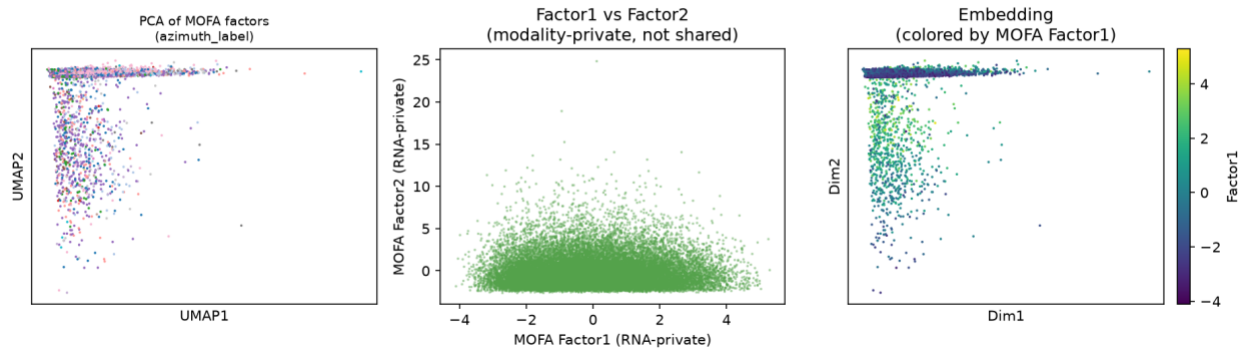
factor	rna	atac_cbg	dominant_view	nondominant_r2	rna_minus_atac
Factor1	0.000244	0.795037	atac_cbg	0.000244	-0.794793
Factor2	0.516283	0.000179	rna	0.000179	0.516104
Factor3	0.171526	0.000263	rna	0.000263	0.171263
Factor4	0.145588	0.000222	rna	0.000222	0.145366
Factor5	0.128056	0.000189	rna	0.000189	0.127867
Factor6	0.11177	0.000194	rna	0.000194	0.111576
Factor7	0.079673	0.000191	rna	0.000191	0.079482
Factor8	0.00035	0.058451	atac_cbg	0.00035	-0.058101
Factor9	0.000415	0.055252	atac_cbg	0.000415	-0.054837
Factor10	0.04313	0.000134	rna	0.000134	0.042996
Factor11	0.03681	0.000115	rna	0.000115	0.036695
Factor12	0.025438	6.39e-05	rna	6.39e-05	0.025374
Factor13	0.025268	6.85e-05	rna	6.85e-05	0.0252
Factor14	0.000127	0.023684	atac_cbg	0.000127	-0.023557

Showing 14 of 18 rows. Source: module3_mofa_factor_dominance.tsv



MOFA variance explained per factor for RNA and ATAC views

Donor_2 MOFA factors (stacked modality-private spaces -- see variance figure)



MOFA latent embedding for teaching inspection

Units first. The `units` column reads `percent`. Stored totals near 1.3 and 0.91 mean about 1.3 percent of RNA variance and about 0.91 percent of ATAC gene-activity variance, not fractions above 1. The recomputation in `outputs/reports/module3_run_params.json` confirms the percent convention.

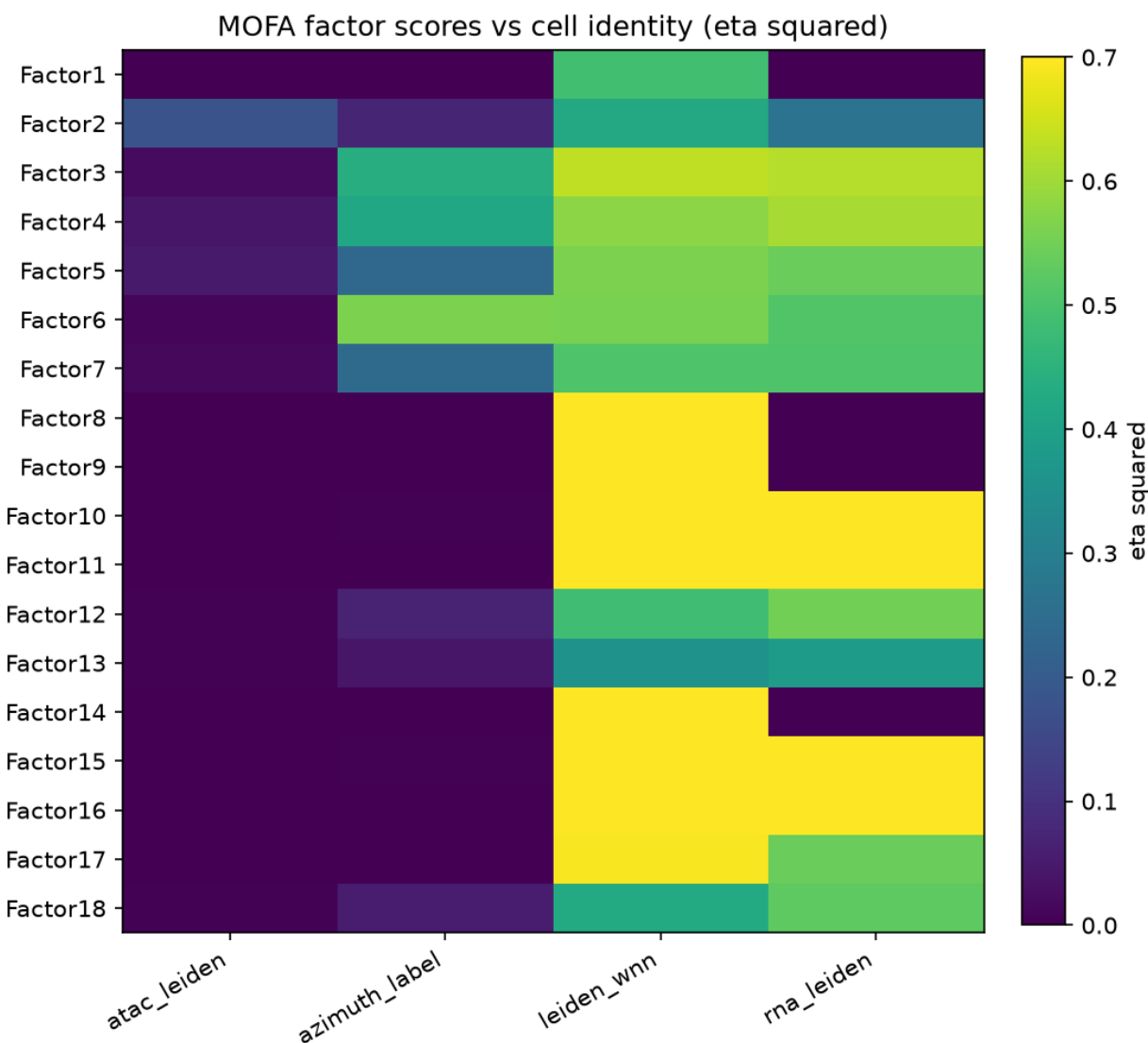
Finding. Every active factor is modality-private. Trailing factors beyond `mofa_variance_max_factor` are numerically dead. This is two stacked single-modality spaces, not a joint latent space.

Factor association with cell identity

factor	labelling	eta_squared	n_groups	n_nuclei
Factor1	azimuth_label	0.001423	34	21978
Factor1	leiden_wnn	0.487874	23	21978
Factor1	rna_leiden	0.000317	20	21978
Factor1	atac_leiden	0.001658	30	21978
Factor2	azimuth_label	0.073788	34	21978
Factor2	leiden_wnn	0.421429	23	21978
Factor2	rna_leiden	0.266424	20	21978
Factor2	atac_leiden	0.178713	30	21978
Factor3	azimuth_label	0.437111	34	21978

Factor3	leiden_wnn	0.632004	23	21978
Factor3	rna_leiden	0.623223	20	21978
Factor3	atac_leiden	0.019384	30	21978
Factor4	azimuth_label	0.416625	34	21978
Factor4	leiden_wnn	0.577355	23	21978

Showing 14 of 120 rows. Source: module3_factor_label_association.tsv



Eta squared of MOFA factor scores against Azimuth, WNN, RNA Leiden, and ATAC Leiden

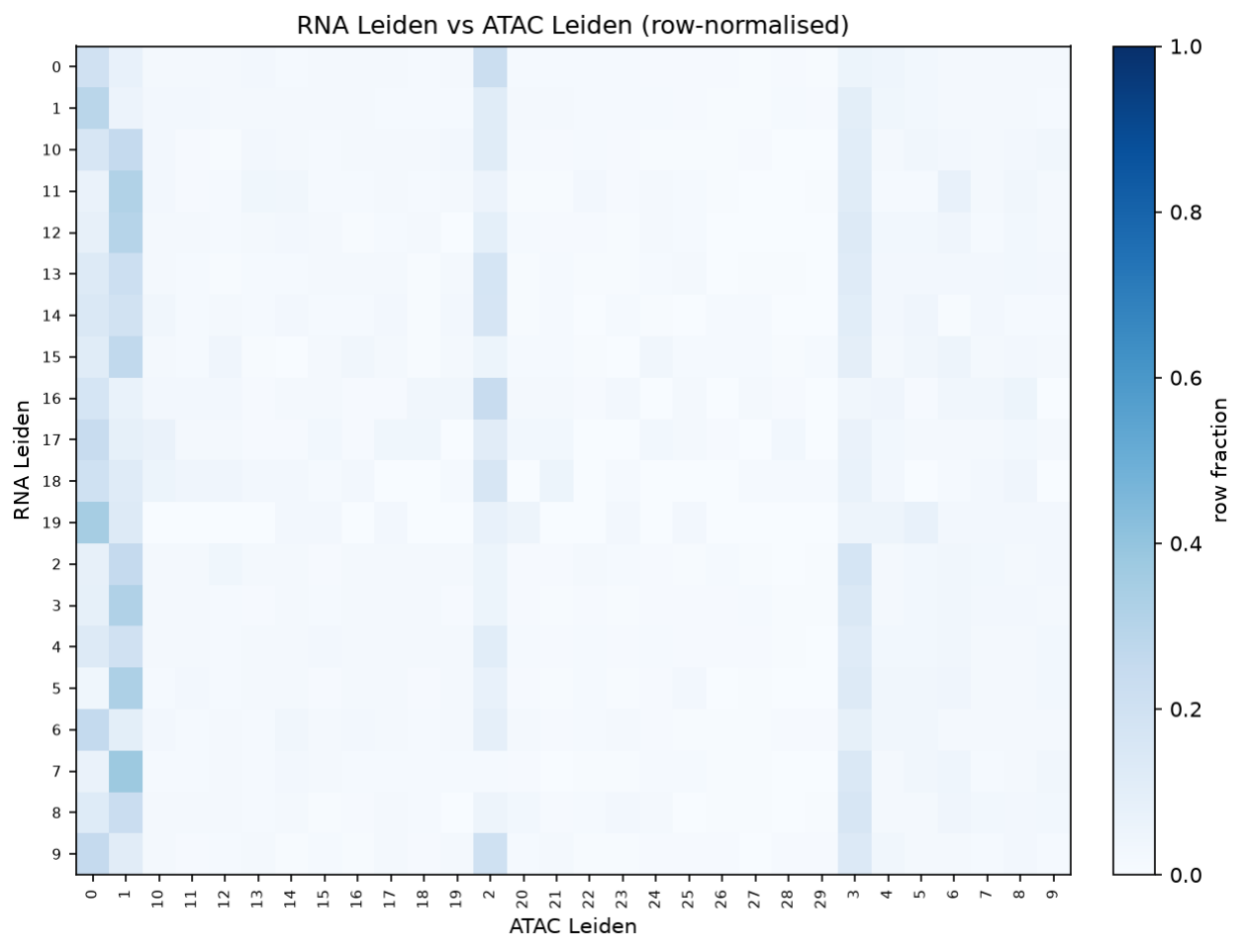
Finding. RNA-dominant factor scores track Azimuth, WNN, and RNA Leiden. The same factors are near floor against ATAC Leiden. MOFA found cell identity, in RNA only.

Content of the ATAC gene-activity view

pair	col_a	col_b	n_nuclei	k_a	k_b	ari	nmi
atac_leiden vs azimuth_label	atac_leiden	azimuth_label	21978	30	34	0.011747	0.028309

atac_clusters vs azimuth_label	atac_clusters	azimuth_label	21978	25	34	0.006943	0.01062
leiden_wnn vs rna_leiden	leiden_wnn	rna_leiden	21978	23	20	0.246892	0.443028
leiden_wnn vs atac_leiden	leiden_wnn	atac_leiden	21978	23	30	0.020134	0.039789
rna_leiden vs azimuth_label	rna_leiden	azimuth_label	21978	20	34	0.180146	0.26221
leiden_wnn vs azimuth_label	leiden_wnn	azimuth_label	21978	23	34	0.065516	0.200672
atac_leiden vs rna_leiden	atac_leiden	rna_leiden	21978	30	20	0.029988	0.040883

1 further column in the source file. Source: module3_cluster_concordance.tsv

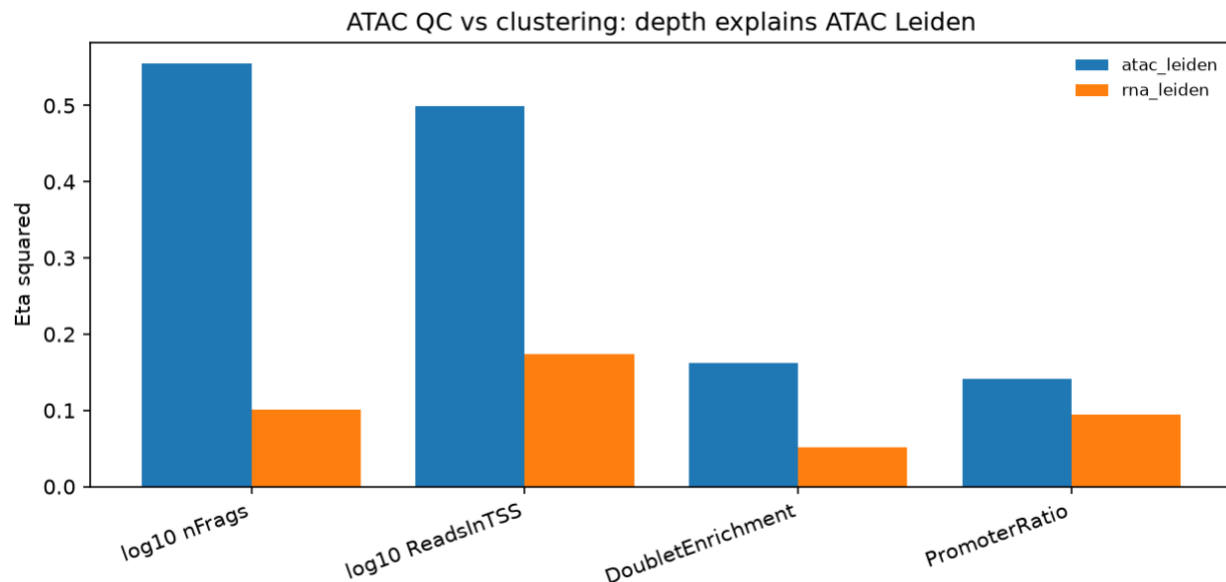


RNA Leiden versus ATAC Leiden, row-normalized

qc_metric	qc_column	labelling	eta_squared	transform	n_nuclei	n_groups
log10 nFrag	nFrag	atac_leiden	0.554629	log10	21978	30
log10 nFrag	nFrag	rna_leiden	0.10157	log10	21978	20
log10 ReadsInTSS	ReadsInTSS	atac_leiden	0.498693	log10	21978	30
log10 ReadsInTSS	ReadsInTSS	rna_leiden	0.174031	log10	21978	20
DoubletEnrichment	DoubletEnrichment	atac_leiden	0.161544	none	21978	30

DoubletEnrichment	DoubletEnrichment	rna_leiden	0.051957	none	21978	20
PromoterRatio	PromoterRatio	atac_leiden	0.141002	none	21978	30
PromoterRatio	PromoterRatio	rna_leiden	0.094058	none	21978	20

Source: `module3_qc_association.tsv`

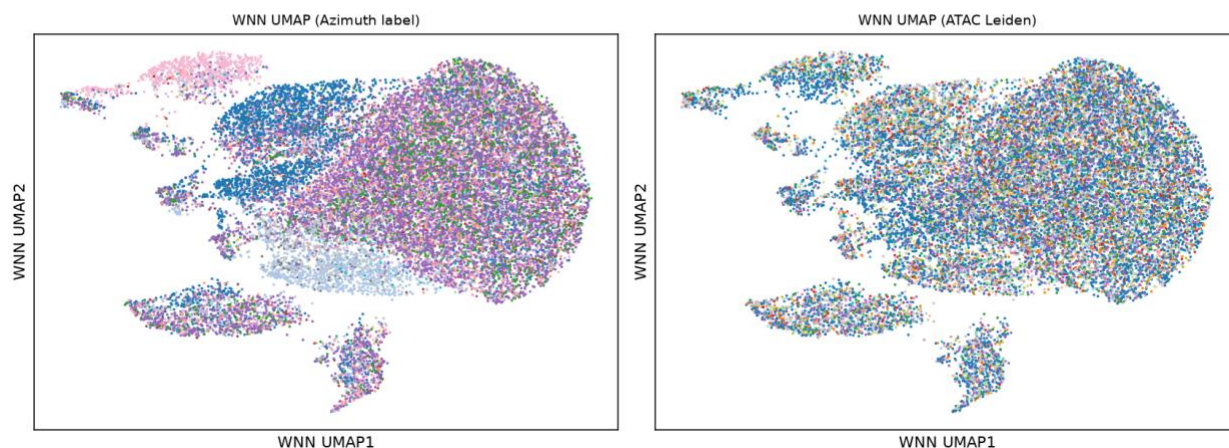


Eta squared of ATAC QC metrics under ATAC Leiden versus RNA Leiden

Finding. ATAC Leiden is near chance against Azimuth cell type. The same clustering tracks sequencing depth (log10 fragment counts and log10 reads in TSS). RNA Leiden does not. On this product the ATAC gene-activity view encodes depth, not cell identity.

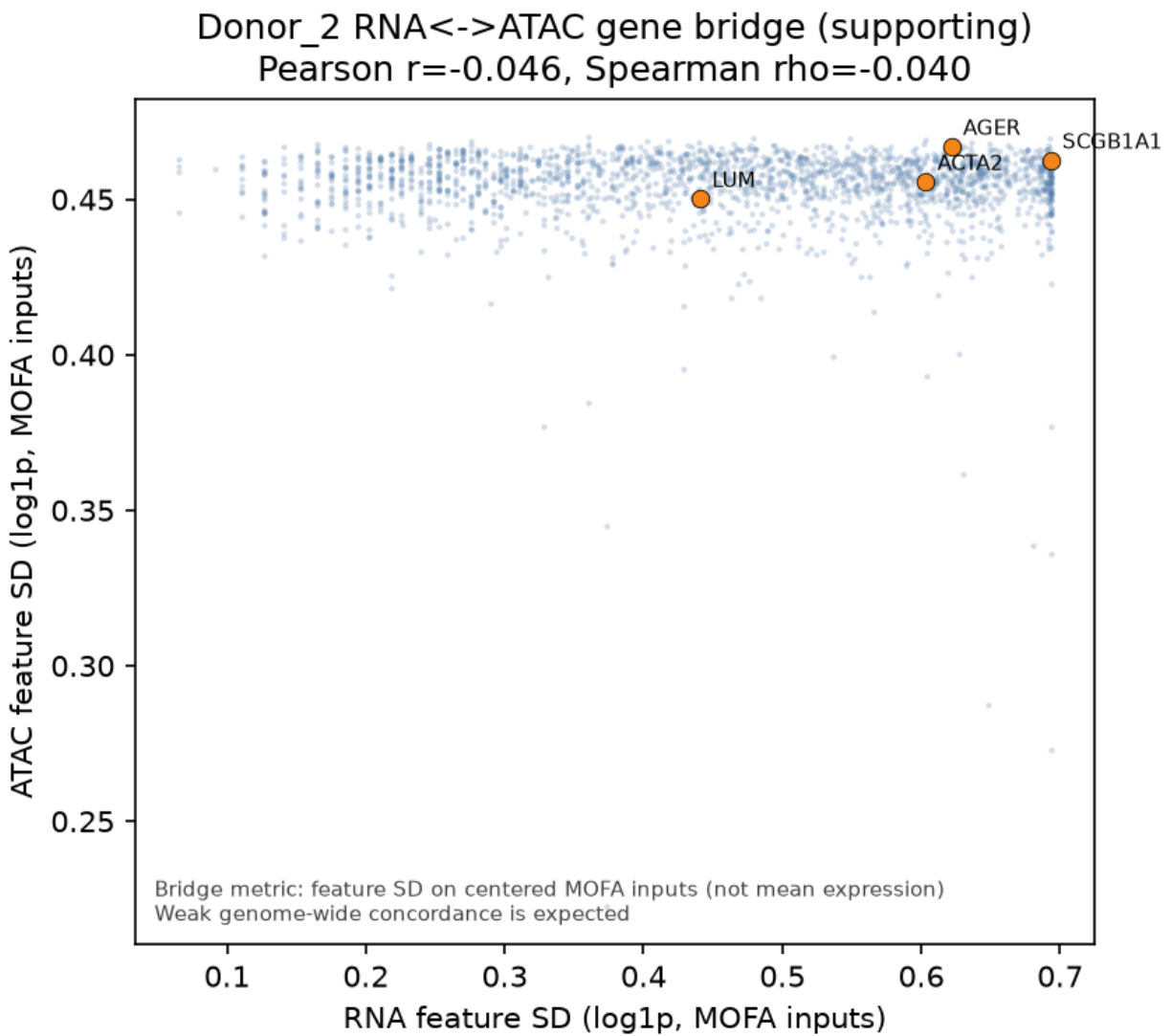
Weighted nearest neighbor concordance

Donor_2 WNN embedding follows RNA identity more than ATAC



WNN UMAP colored by Azimuth label and by ATAC Leiden

Read the WNN rows of `module3_cluster_concordance.tsv`. WNN agrees with RNA Leiden more than with ATAC Leiden, and it agrees with Azimuth less than RNA Leiden does. The joint embedding that looks successful is leaning on RNA.



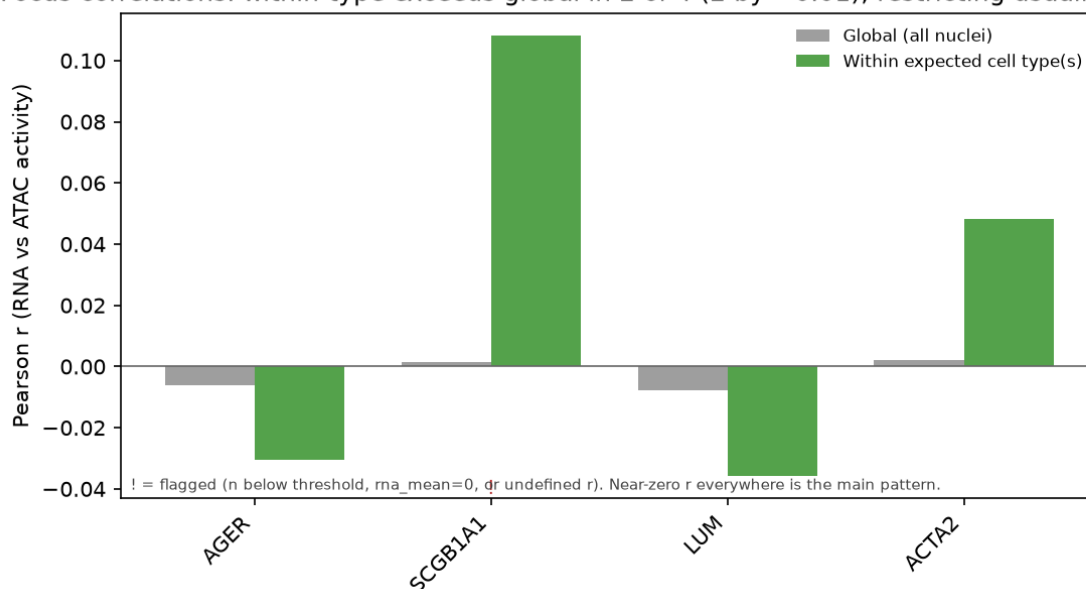
Scatter of RNA vs ATAC feature dispersion for shared genes in the MOFA input space

gene_symbol	scope	n_nuclei	pearson_r	note
AGER	global	21978	-0.006238	
SCGB1A1	global	21978	0.001598	
LUM	global	21978	-0.007713	
ACTA2	global	21978	0.002102	
AGER	within_cell_type	2210	-0.030638	
SCGB1A1	within_cell_type	37	0.108276	n_nuclei=37 below threshold 50; correlation not interpretable
LUM	within_cell_type	1750	-0.035931	

ACTA2	within_cell_type	343	0.04819	
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5 further columns in the source file. Source: `module3_focus_gene_correlations.tsv`

Focus correlations: within-type exceeds global in 2 of 4 (2 by >0.01); restricting usually shrinks r



Focus-gene RNA vs ATAC correlations, global and within cell type

Shared HUGO symbols and genome-wide correlations are in `module3_rna_atac_gene_bridge.tsv` (row count) and its figure caption. Within-cell-type rows are now populated because the cache supplies Azimuth labels. Low-n groups are flagged rather than reported as bare missing values.

5. Interpretation

Module 3 teaches how to interrogate a real multiome product without forcing a shared-factor story the object does not support.

This object is a legitimate multiome: paired RNA and ATAC in the same nuclei, with a matched 10x snRNA block from the same section. The ATAC **gene-activity** representation is not a legitimate basis for integration on this product. Those two statements are not in conflict.

What the results support.

- MOFA did not fail as a method. RNA factors track cell identity.
- The ATAC gene-activity view carries no detectable cell-type information, and its clustering tracks sequencing depth.
- WNN produces a joint embedding by following RNA, and it matches Azimuth less well than RNA alone.
- Factor dominance and dead factors remain first-class results: they constrain claims about joint latent biology.

Success here means the files pair, the factor model loads, the variance tables are interpretable, and the diagnostics distinguish method failure from empty ATAC structure. It does **not** mean RNA and ATAC share a latent axis. If a manuscript claimed shared MOFA factors for this object without showing dual-view R2 and a cell-type check, Module 3 is the counterexample.

Two limits to name explicitly. Azimuth labels are transferred predictions rather than ground truth. WNN and Azimuth diagnostics are measured on Donor_2 only.

6. Extending this analysis

Each item below names a claim a learner may want to make and the work that would license it.

- **Whether the depth confound is block-specific.** [HBM285.MW NX.593](#) is a Donor_3 multiome from a different block, ten libraries against ten. Repeating the QC association there would test whether ATAC gene-activity clustering is depth-driven outside Donor_2.
- **Peak-level ATAC.** Differentially accessible regions, motif enrichment, and chromVAR need peak-level data. The shipped ATAC view is gene activity, and that tooling is not in the pinned environment.
- **The bin-level modality.** [atac_cbb](#) is present in the MuData object with chromosome and bin coordinates. It is not in the MOFA object and is not on the teaching path ([use_atac_bin](#): false).
- **Refitting MOFA.** The lesson reads a DCC fit ([interpretation_only](#): true). A refit changes the object under discussion and cannot be reproduced identically across environments.
- **Independent pairing evidence.** A barcode crosstab from MuData [obs](#) would confirm shared nuclei independently. The default path asserts pairing from MOFA file construction.
- **Donor_3 as a second arm.** Stated as an extension, not run. Donor_3 cannot support a matched snRNA/multiome comparison at any download size, because none of its 10x blocks share a block with its SNARE products.

Forward link. Module 4 leaves single-DCC multiome and compares bulk and snRNA-derived contrasts across HuBMAP, GTEx, GEO, and OSDR.

7. Open questions

1. What evidence shows this is true multiome rather than unpaired same-donor assays, and what does the matched 10x block license that Donor_3 could not?
2. Why is this MOFA space not a joint latent space? Quote the max non-dominant R2 from the table.
3. What result shows that MOFA still found biology? Quote an eta squared for an RNA factor against Azimuth.
4. Why is the ATAC gene-activity view not a legitimate basis for integration on this object?
5. WNN looks like a joint embedding. What comparison shows that it is leaning on RNA?
6. When a within-cell-type focus-gene row is flagged for low n, what claim is still licensed?
7. Factor1 carries the largest single share of ATAC variance yet tracks no cell-type label, no cluster assignment and no recorded QC metric, while sitting at eta squared 0.488 against the WNN clustering. WNN takes the ATAC representation as an input. What does that shared input explain, and what does it leave open?

Appendix A: Analysis parameters

Live config values under `module3` in `config/paths.yaml`, in Methods order.

1. Default path: interpret shipped MOFA

Config key	Default	Definition	If modified	Impact on results
<code>module3.params.load_h5mu</code>	false	Load full secondary_analysis.h5mu	Set true	Enables MuData UMAP/layers; large RAM/download; not required
<code>module3.params.interpretation_only</code>	true	Skip MOFA refit; read shipped factors	Set false	Out of scope for class; non-identical refits
<code>module3.files.mofa_hdf5</code>	multiome_mofa.hdf5	Filename under multiome dir	Point elsewhere	Different factor object
<code>module3.multiome_dir</code>	data/source/HUBMAP/Donor_2/SNAREseq/HBM828.GPVG.252	Local product directory	Relocate	Path resolution
<code>module3.donor_label</code>	Donor_2	Teaching donor	Change donor	Every Module 3 claim is donor-specific
<code>module3.dataset_id</code>	HBM828.GPVG.252	Public SNARE product ID (folder name)	Change product	Different object
<code>module3.modalities.rna</code>	rna	MOFA/MuData RNA view name	Rename	Load fails if view missing
<code>module3.modalities.atac_gene</code>	atac_cbg	ATAC gene-activity view	Switch	Bridge uses different ATAC features
<code>module3.modalities.use_atac_bin</code>	false	Whether to use atac_cbb	Set true	Adds bin-level ATAC path

2. Barcode pairing check

Config key	Default	Definition	If modified	Impact on results
<code>module3.params.load_h5mu</code>	false	Whether independent MuData barcode sets are available	Set true	Overlap can be recomputed from MuData obs
<code>module3.files.secondary_h5mu</code>	secondary_analysis.h5mu	MuData filename when loaded	Missing file	Blocks MuData path only

3. Variance explained and factor dominance

Config key	Default	Definition	If modified	Impact on results
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module3.analysis.mofa_variance_max_factor	18	Last factor treated as active for teaching	Raise	May present near-zero factors as active
module3.analysis.random_seed	0	Seed for any stochastic preview draws	Change	Preview subsample only
module3.analysis.max_nuclei_preview	5000	Cap when MuData preview embeddings are drawn	Raise	Higher RAM on MuData path

4. Factor association with cell identity

Config key	Default	Definition	If modified	Impact on results
module3.files.label_cache	outputs/tables/module3_label_cache_HBM828.GPVG.252.tsv	Committed Azimuth / WNN cache	Rebuild cache	All labeled diagnostics change
module3.thresholds.min_label_prediction_score	0.0	Azimuth score floor (off)	Raise	Drops low-confidence labels

5. Content of the ATAC gene-activity view

Config key	Default	Definition	If modified	Impact on results
module3.thresholds.min_cluster_size	20	Recorded cluster-size floor	Raise	Does not change default ARI; reserved for extensions

6. Weighted nearest neighbor as a second integration

Config key	Default	Definition	If modified	Impact on results
module3.files.label_cache	see section 4	Supplies leiden_wnn and WNN UMAP	Rebuild	WNN comparisons and UMAP change
module3.matched_snrna.primary_id	HBM473.NKMR.872	Matched 10x snRNA teaching block	Change	Cross-assay language only; no barcode join

7. Genome-wide RNA-ATAC gene bridge

Config key	Default	Definition	If modified	Impact on results
module3.analysis.bridge_metric	feature_std_on_mofa_inputs	Summary used for genome-wide scatter	Change metric	Correlation magnitudes redefine
module3.analysis.rna_layer	spliced_unspliced_sum	Preferred MuData RNA layer	Use spliced only	Layer-resolved values change on MuData path
module3.analysis.rna_layer_fallback	spliced	Fallback if preferred layer absent	N/A if preferred exists	Contingency only
module3.thresholds.noise_delta_r	0.01	Within-type vs global margin treated as noise	Raise	More correlations flagged as noise

8. Focus-gene correlations and low-n guards

Config key	Default	Definition	If modified	Impact on results
module3.analysis.focus_genes	SFTPC, AGER, ...	Genes plotted/correlated	Edit list	Which rows appear; not genome-wide stats
module3.analysis.focus_gene_cell_types	per-gene Azimuth lists	Expected cell types for within-type n	Edit	Changes within-type n and flags
module3.analysis.focus_genes_not_in_both_modalities	[EPCAM]	Genes known unpaired	Edit	Documentation of impossible pairs
module3.thresholds.min_nuclei_correlation	50	Flag low-n correlations	Raise	More rows flagged unreliable
module3.thresholds.noise_delta_r	0.01	Noise margin vs global r	Raise	Stricter "real within-type" calls

Optional MuData extension

Config key	Default	Definition	If modified	Impact on results
module3.params.load_h5mu	false	Whether MuData extension runs	Set true	Large download/RAM; extra panels
module3.params.secondary_h5mu_size_note	~14.7 GiB; optional...	Human-readable size warning	Editorial	Docs only
module3.interpretation	true	Write interpretation markdown	Set false	Skips narrative export
module3.assay	SNARE-seq2 [Salmon + ArchR + Muon]	Assay label for reports	Editorial	Provenance text

References

- Duong TE, Diep D, Conklin KY, et al. A Multimodal Spatial and Epigenomic Atlas of Human Adult Lung Topography. *bioRxiv*. 2025. DOI: [10.1101/2025.05.23.655666](https://doi.org/10.1101/2025.05.23.655666). Preprint; the source publication for the SNARE-seq2 and snRNA-seq data this course uses. Its own analysis labels SNARE-seq2 nuclei from the RNA modality alone and reports no joint RNA-ATAC factor model, so the MOFA object interpreted here is a product of the HuBMAP central pipeline rather than of that paper.
- Argelaguet R, et al. MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome Biol*. 2020;21(1):111. PMID: [32393329](https://pubmed.ncbi.nlm.nih.gov/32393329/).
- HuBMAP Consortium. The human body at cellular resolution: the NIH Human Biomolecular Atlas Program. *Nature*. 2019;574(7777):187-192. PMID: [31597973](https://pubmed.ncbi.nlm.nih.gov/31597973/).
- Dataset: HuBMAP [HBM828.GPVG.252](#) (SNARE-seq2 Salmon + ArchR + Muon). Portal: [browse dataset](#).
- Matched 10x snRNA teaching block: HuBMAP [HBM473.NKMR.872](#). Portal: [browse dataset](#).